

Research Paper Explanation

Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites

Based on Mathur et al. (2019)

Paper focus

This study investigates manipulative interface designs on shopping websites. It examines how common dark patterns are, how they work, and how they may harm users through deception, pressure, hidden information, and forced actions.

1. Title

Meaning of the title: The paper studies dark patterns across a very large sample of shopping websites rather than relying on a few examples.

- Dark patterns: interface choices that deceive, pressure, or steer users into decisions they may not otherwise make.
- At scale: the researchers used automated tools to analyse thousands of websites.
- 11K shopping websites: the study focuses on product, cart, and checkout experiences in online stores.

Simple interpretation

The paper measures how widespread manipulative shopping interfaces are and explains the mechanisms through which they influence users.

2. Abstract

The researchers developed automated techniques to identify dark patterns on approximately 11,000 shopping websites. They analysed around 53,000 product pages and found:

- 1,818 dark-pattern instances.
- 15 different types grouped into 7 broader categories.
- Dark patterns on 1,254 websites, approximately 11.1% of the sample.
- Deceptive practices on 183 websites.
- 22 third-party companies that supplied tools capable of enabling dark patterns.

Main message: Dark patterns are not rare accidents. They form a broader commercial ecosystem involving retailers, interface designers, and third-party technology providers.

3. Introduction: First 1-2 Paragraphs

The introduction defines dark patterns as interface choices that benefit a company by coercing, steering, or deceiving users into decisions they might not make if they were fully informed and able to choose freely.

These designs may cause unwanted purchases, hidden subscriptions, financial loss, unnecessary data disclosure, and compulsive behaviour. Previous research had already classified dark patterns, but there was limited large-scale evidence about how often they appear, where they appear, and how they harm users.

Research gap: The paper fills this gap by using automated website crawling and systematic classification to document dark patterns across a large sample of online stores.

4. Conclusion

The authors conclude that automated systems can identify dark patterns across large numbers of websites and can support researchers, regulators, journalists, and consumer-protection tools.

- At least 11.1% of the examined shopping websites contained dark patterns.
- More popular websites were more likely to contain them.
- Many patterns relied on deception, hidden information, psychological pressure, and unequal choices.
- Third-party providers made these practices easier for websites to deploy.
- Browser warnings, stronger regulation, public awareness, and further behavioural studies are needed.

Important caution: The reported 11.1% is a lower-bound estimate because the crawler focused mainly on text-based patterns and selected shopping pages. It could not fully detect manipulation based on colour, layout, imagery, or post-purchase interactions.

Three Questions to Ask Yourself

What problem are they solving?

They address the lack of large-scale evidence about how shopping websites use manipulative interface designs to influence purchases, subscriptions, and disclosure decisions.

Why is it important?

Dark patterns weaken independent decision-making and may cause unnecessary spending, hidden recurring payments, unwanted data disclosure, difficult cancellation, and decisions based on false urgency or scarcity.

What did they find?

Dark patterns are widespread, more common on popular websites, often deceptive or information-hiding, and supported by third-party companies that make manipulation easier to deploy.

Applying the Paper to Five Ethical Questions

How much user data should a web application collect?

The paper does not prescribe an exact quantity, but its findings support data minimisation. An application should collect only data that is necessary for a clearly defined function, proportionate to the service, explained to the user, and retained only as long as required.

The paper treats forced enrolment - requiring account creation or unnecessary information before completing a task - as a dark pattern.

Is user consent alone sufficient to make data collection ethical?

No. Consent is not ethically meaningful when it is obtained through confusing language, preselected choices, emotional pressure, hidden consequences, unequal options, or forced registration. Ethical collection also requires necessity, transparency, fairness, security, and meaningful control.

How do dark patterns influence users' privacy decisions?

They exploit predictable cognitive biases:

- Default effect: users tend to keep preselected options.
- Framing effect: wording or presentation changes how a choice is perceived.
- Scarcity bias: users act quickly when opportunities appear limited.
- Bandwagon effect: users follow what others appear to be doing.
- Sunk-cost fallacy: users continue after investing time or effort.
- Anchoring effect: initial information shapes later judgement.

Thus, a user may appear to choose freely even though the interface has already shaped the decision.

What responsibilities do software engineers have?

- Avoid deceptive defaults, misleading messages, and hidden consequences.
- Make acceptance and rejection equally understandable and accessible.
- Disclose fees and recurring subscriptions before commitment.
- Avoid fake scarcity, false countdowns, and fabricated activity messages.
- Collect only necessary personal data.
- Make cancellation as easy as registration.
- Challenge unethical product requirements rather than treating them as neutral business requests.
- Test whether interfaces preserve user autonomy, not only whether they increase conversion.

How can companies balance personalization and user privacy?

1. Provide the basic service without unnecessary profiling.
2. Explain what data improves personalization and why it is needed.
3. Request optional information separately.
4. Allow refusal without removing essential functionality.
5. Prefer first-party, aggregated, or on-device data where possible.
6. Allow users to review, change, and delete their preferences and data.

Core principle: Personalization should offer clear value to the user, not justify unlimited data collection.

5. Research Questions

The paper does not state formally numbered research questions, but its objectives can be expressed as follows:

1. How prevalent are dark patterns on popular shopping websites?
2. What types and categories of dark patterns are used?
3. How do dark patterns influence or harm user decision-making?
4. Which cognitive biases do these designs exploit?
5. Do third-party technology providers enable websites to deploy dark patterns?
6. Can automated techniques identify dark patterns at a large scale?

6. Methodology

The study used a large-scale computational and observational method.

Website selection

The researchers began with approximately 361,000 websites from Alexa rankings. After classifying shopping sites and filtering for English-language content, they identified 19,455 shopping websites.

Automated crawling

They developed two crawlers:

- A product-page crawler that identified product pages.
- A checkout crawler that simulated selecting products, adding them to the cart, and moving toward checkout.

The system collected 53,180 product pages from 11,286 websites.

Data collection and analysis

- HTML source code, screenshots, network records, interface text, and page changes were collected.
- Approximately 13 million interface segments were extracted.
- Duplicates were removed and text was converted into a bag-of-words representation.
- PCA reduced dimensionality, and HDBSCAN clustering organised similar interface segments.
- Researchers manually examined relevant clusters and classified the patterns.
- Dynamic patterns such as countdown timers were revisited every four hours for five days to test whether they were deceptive.

7. Results

The researchers found 1,818 dark-pattern instances across 1,254 websites.

Category	Meaning
Sneaking	Adds items, fees, or subscriptions without proper disclosure
Urgency	Uses countdown timers or limited-time messages to accelerate decisions
Misdirection	Uses language, emotion, or visual design to steer users
Social proof	Claims that other people are viewing or buying products
Scarcity	Uses low-stock or high-demand messages
Obstruction	Makes actions such as cancellation unnecessarily difficult
Forced action	Requires registration or information disclosure to continue

Major examples included:

- 632 low-stock messages.
- 393 countdown timers.
- 313 activity messages.
- 169 confirmshaming examples.
- 88 limited-time messages.
- 67 pressured-selling examples.
- 31 hard-to-cancel systems.

The study identified 234 deceptive dark-pattern instances across 183 websites and found 22 third-party companies offering technologies that enabled such designs.

8. Discussion

The findings show that dark patterns are part of a broader business ecosystem, not merely isolated design mistakes.

Implications for users

- Browser extensions could identify suspicious patterns.
- Warning symbols could flag deceptive content.
- Public scoring systems could rate websites according to manipulative practices.
- Community-maintained lists could work similarly to ad-blocking systems.

Implications for companies and engineers

Short-term conversion gains obtained through deception may create long-term losses in trust, legal exposure, and customer loyalty. Engineers and product teams should evaluate whether a design preserves informed choice rather than merely measuring whether it increases sales or sign-ups.

Implications for regulators

Automated monitoring can help regulators investigate dark patterns at scale. Enforcement should address both websites that deploy manipulative interfaces and third-party vendors that supply these tools.

Limitations

- The study identified and classified patterns but did not directly measure their effect on users through an experiment.
- It focused on English-language shopping websites.
- It mainly examined text-based interfaces on product, cart, and checkout pages.
- It did not fully observe post-purchase experiences or purely visual manipulation.

Overall interpretation: The true prevalence of dark patterns is likely higher than the reported 11.1%, but the study still provides strong evidence that manipulative design is systematic and measurable.

Final takeaway

A well-designed system should help users make informed decisions. When an interface hides information, creates false urgency, forces disclosure, or makes refusal difficult, it stops supporting the user and begins manipulating the user.

Source paper: Mathur, A. et al. (2019). Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites.